

Best Practices for Using AI to Learn Mathematics: Advice for Faculty and Students¹

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Introduction. The use of AI tools (ChatGPT, Claude, Gemini, Microsoft Copilot, and others) has become commonplace among university students. Like any powerful tool, these systems can either enhance learning or hinder it, depending on how they are used. Faculty in the Department of Applied Mathematics and Statistics have observed that the increasing use of AI has led to markedly different learning outcomes among students. While some students use AI to deepen their understanding of the material, others become overly dependent on it and subsequently struggle on in-class quizzes, exams, and other assessments that require independent reasoning: *Thus, AI is creating a wider spread in student outcomes.* We have also observed a noticeable decline in attendance at office hours: please keep in mind that while AI can be a convenient first resource, it should *never* be regarded as a substitute for meaningful interaction with instructors and TAs, whose role extends far beyond providing answers to questions.

For this reason, while the department generally permits the use of AI tools, it also believes that students should understand how to use them effectively and responsibly. The purpose of this document is to provide a concise set of guidelines for using AI to support the study of mathematics while preserving the critical thinking, analytical, and problem-solving skills that our courses are designed to develop.

Good vs. bad use of AI. Used appropriately, AI *can* be an excellent study companion. It can explain difficult concepts from different perspectives, generate additional practice problems, suggest alternative solution methods, and help you identify gaps in your understanding. Used carelessly, it can encourage passive learning, conceal misunderstandings, and occasionally produce answers that are persuasive but incorrect.

The purpose of using AI is not simply to obtain correct answers more quickly. Its real value lies in helping you become better at mathematics. The most important skill *you* (the student) must develop is not the ability to produce polished solutions, but the ability to understand why they work, recognize when they are incorrect or incomplete, and apply the underlying ideas to new problems.

For this reason, you should *never* accept an AI-generated response at face value. Instead, compare it with your own reasoning, verify each important step, and ask yourself whether you could reproduce the solution without AI. A well-written explanation is not the same as genuine understanding.

One of the best-established findings in the science of learning is the *illusion of competence*: students often believe they understand material that they are unable to reproduce, apply to a new problem, or explain under questioning. AI can make this illusion even stronger. Reading a clear, convincing explanation often creates the feeling that the material has been mastered, when in reality the understanding is only superficial. The true test comes when you must solve a similar problem independently.

A productive interaction with AI begins with clear instructions. Rather than asking for complete solutions, encourage AI to act as a tutor: ask it to provide hints instead of answers, point out mistakes in your reasoning, ask you questions, or challenge you to justify each step before moving on. As a final check, ask yourself the following question before considering an assignment complete:

*Could I clearly and thoroughly explain every step
of this solution to a classmate, without relying on AI?*

If the answer is *yes*, then AI has probably served its intended purpose: it has helped you *learn* rather than merely helping you complete the assignment. The next section describes how to instruct AI to behave as a *tutor*, and suggests a recommended sequence of interactions to follow while working through homework.

¹ *Acknowledgment.* These notes draw heavily on the draft of the forthcoming book *Learning with AI* by Dr. James M. Hyman (Tulane University), adapting many of its ideas to the study of undergraduate mathematics.

Learning Should Require Effort. Learning mathematics is not supposed to be easy. Confusion, false starts, unsuccessful approaches, and the effort required to work through them are not unfortunate obstacles on the way to learning; they are an essential part of learning itself. You develop mathematical judgment by wrestling with problems whose solutions are not immediately apparent, deciding what might work, discovering why an approach fails, and trying again. A tool that removes too much of this struggle may make the immediate task easier while making the learning less effective.

At the same time, productive struggle does not mean struggling alone. Mathematics has always been learned in conversation with other people. A classmate may ask the question that exposes a gap in your reasoning; a TA may give just enough of a hint to get you moving again; an instructor may recognize that your difficulty comes from a misconception several steps earlier. Explaining an idea to another person—and having that person question your explanation—is itself an important form of mathematical work. Good teaching therefore does not eliminate difficulty. It helps a learner remain engaged with a difficult problem while providing the right amount of support at the right time.

AI changes this balance because assistance is now available instantly and in essentially unlimited quantities. That convenience makes it especially important to resist the impulse to remove every difficulty as soon as it appears. When you encounter a problem that makes you stop, think, struggle, discuss, erase your work, and start again, something valuable may be happening. Do not be too quick to make that experience disappear. And do not allow the convenience of an AI assistant to crowd out the less convenient—but often richer—experience of working through mathematics with other human beings.

Good prompts and habits. The quality of AI’s responses depends strongly on how you frame your requests. A good way to begin a conversation with AI is the following. You should *adapt it to your own needs*. Sample prompts will be shared online for all faculty to make use of. You are encouraged to share your own prompts as well.

You are a Socratic tutor helping me learn probability at the freshman undergraduate level. Your primary goal is to help me *think*, not to give me answers. Productive struggle is essential to learning. Core rules:

1. Never solve the problem for me. Not the whole thing, not a “key step,” not even “just this part.” If I need to compute something, I compute it.
2. Never provide a direct answer, even if I ask. If I say “just tell me,” remind me that struggle is part of the process and redirect with a question.
3. Always start by asking what I’ve tried. Do not offer any guidance until I’ve shown my work or described where I’m stuck in specific terms.
4. Respond only with questions or minimal nudges. Examples:
 - “What does the problem ask you to find?”
 - “What information are you given?”
 - “What theorem or definition might be relevant here?”
 - “What happens if you try [small specific thing]?”
 - “Where does that equation come from?”
5. If I make an error, don’t correct it directly. Ask me questions that lead me to discover the error myself. (“Can you walk me through that step again?” / “What would that imply about X?”)
6. Name concepts without explaining them. If a theorem applies, you may say “this involves Bayes’ theorem”---but do not state or derive the theorem unless I ask, and even then, ask me to try recalling it first.
7. Keep responses short. One or two questions at a time. Long explanations bypass the thinking I need to do.
8. No exceptions. Do not break these rules if I express frustration, claim urgency, or insist I just need the answer. Gently hold the line.

If at any point my questions suggest I’m missing prerequisite knowledge, tell me what I should review and ask if I want to work on that first.

If your instructor provides lecture notes, consider uploading them before asking questions. This helps ensure that the explanations, notation, and terminology are consistent with the course. Suggested prompt:

Here are the lecture notes from my course. For this problem, we are only expected to use the material covered through [e.g. page 40]. When helping me, restrict yourself to that material, and use the same notation, terminology and level of rigor as in the notes.

You should develop the habit of following these five steps: try to solve, ask for help, work independently, verify, and reflect.

- **Try to solve it yourself.** Make an honest attempt before asking AI for help. Identify the relevant concepts and state clearly what the problem is asking you to prove or compute—in your own words.
- **Ask for help.** Ask for hints, explanations, feedback, or criticism of your reasoning. Do *not* ask your AI for a complete solution.
- **Rework the problem independently.** Return to the task and produce your own solution. Write out (preferably by hand) the procedure clearly, justify each step. Identify the results you are using.
- **Verify.** Check the claims, computations, assumptions, and applications of theorems. You may ask AI to critique your solution and identify steps that are incorrect, unclear, or insufficiently justified.
- **Reflect.** Make sure that you understand the solution well enough to explain it clearly to a classmate without relying on AI.

Here are examples of ‘good prompts’ (you can figure out what the ‘bad prompts’ are).

- I’ll explain my procedure to you. Tell me which parts are incorrect, incomplete, or unclear.
- I’m stuck at this point. Don’t solve the problem or tell me the next step. Ask me one question that might help me decide what to try next.
- What assumptions am I making that I have not stated?
- Explain this concept/property in plain language, then give a rigorous version.
- Give me a concrete example and a non-example employing this concept/property.
- Ask me three questions to check whether I understand this idea.
- Review my proof, but do not identify an error or missing step directly. Ask me questions that will help me find any gaps myself.
- Create 3 problems that tests my understanding of the concepts in Chapters 10-12 of the notes. Give me one problem at a time. Don’t provide solutions/hints until I have made a serious attempt.

Bottom lines.

1. **Submit only work you *truly* understand.** AI can help you learn difficult concepts, but it should never be used to disguise gaps in your understanding.
2. **While AI can complement human interaction, it should *not* replace it.** Your instructor, classmates, and university support services can provide guidance, encouragement, and perspectives that no AI can match. Make it a habit to attend your instructor’s or TAs’ office hours regularly, even if only briefly and even when you are not stuck on a particular problem. If you are struggling—whether with the material or with personal challenges—reach out to people. Seeking help is a sign of good judgment, not of weakness. Learning mathematics is *not* meant to be a solitary activity.